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Inventor(s)

OKADA; Kohei et al.

POWER GENERATION SYSTEM AND FLIGHT VEHICLE

Abstract

Provided is a power generation system including a photovoltaic unit arranged in a flight vehicle which provides a wireless communication service to a user terminal in a communication area formed over a ground, an electric storage unit which is connected to the photovoltaic unit and stores electric power generated by the photovoltaic unit, and a power conditioner which converts the electric power generated by the photovoltaic unit. The electric storage unit is connected, for example, in parallel to connection between the photovoltaic unit and the power conditioner, and the electric power stored in the electric storage unit is supplied to the power conditioner.

Inventors: OKADA; Kohei (Tokyo, JP), UEDA; Marika (Tokyo, JP)

Applicant: SoftBank Corp. (Tokyo, JP)

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Background/Summary

[0001] The contents of the following patent application(s) are incorporated herein by reference: NO. 2022-193171 filed in JP on Dec. 1, 2022 NO. PCT/JP2023/040773 filed in WO on Nov. 13, 2023.

BACKGROUND

1. Technical Field

[0002] The present invention relates to a power generation system and a flight vehicle.

2. Related Art

[0003] Patent document 1 describes a technique for increasing a recovery rate or a utilization rate of generated power of a power generation apparatus such as a solar power generation panel.

PRIOR ART DOCUMENTS

Patent Document

[0004] Patent Document 1: Japanese Patent Application Publication No. 2020-090852

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 schematically illustrates an example of an HAPS **200**.

[0006] FIG. 2 is an explanatory diagram for describing a power generation system **20** included in the HAPS **200**.

[0007] FIG. 3 schematically illustrates an example of an arrangement of an electric storage unit **234**.

[0008] FIG. 4 schematically illustrates an example of the arrangement of the electric storage unit **234**.

[0009] FIG. 5 schematically illustrates an example of the arrangement of the electric storage unit **234**.

[0010] FIG. 6 schematically illustrates an example of the arrangement of the electric storage unit **234**.

[0011] FIG. 7 schematically illustrates an example of the arrangement of the electric storage unit **234**.

[0012] FIG. 8 schematically illustrates an example of the arrangement of the electric storage unit **234**.

[0013] FIG. 9 schematically illustrates an example of the arrangement of the electric storage unit **234**.

[0014] FIG. 10 schematically illustrates an example of the arrangement of the electric storage unit **234**.

[0015] FIG. 11 schematically illustrates an example of a hardware configuration of a computer **1200** which functions as a control apparatus including a switch control unit **237** and an information acquisition unit **238**.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0016] Since a converter directly connected to a solar cell needs input electric power at or above a certain level, in bad weather or when brightness is not at certain level, the converter cannot efficiently convert weak electric power in many cases, and power generation may be stopped. In a stratospheric platform such as an HAPS (High Altitude Platform Station) to which a solar power

generation panel is mounted and which flies in a stratosphere and forms a wireless communication area over a ground, weak electric power is likely to be generated due to an influence from a shape or a bearing of an airframe or a balloon, a time of day (how the sun shines), or the like, and an amount of wasted electric power is expected to be large. According to the present embodiment, an electric storage unit which recovers weak electric power is introduced to a power generation system of the stratospheric platform.

[0017] Hereinafter, the present invention will be described through embodiments of the invention, but the following embodiments do not limit the invention according to claims. In addition, not all of the combinations of features described in the embodiments are essential to the solution of the invention.

[0018] FIG. 1 schematically illustrates an example of an HAPS **200**. The HAPS **200** may be an example of a flight vehicle which provides a wireless communication service to a user terminal **70** in a communication area **204** formed by emitting a beam **202** towards the ground.

[0019] The HAPS **200** includes a wing portion **220**, a photovoltaic unit **230**, a propeller **240**, an elevator **250**, a central portion **260**, and a pod **270**.

[0020] The photovoltaic unit **230** includes a photovoltaic panel which receives light to generate electric power. The photovoltaic panel may be so-called a solar power generation panel. The photovoltaic unit **230** may be constituted by a plurality of photovoltaic panels. The photovoltaic unit **230** may include a plurality of strings in which a plurality of photovoltaic panels are connected.

[0021] The HAPS **200** includes a plurality of batteries **280** which are not illustrated in the drawing. The plurality of batteries **280** are distributed to be arranged in all or some of the wing portion **220**, the central portion **260**, and the pod **270**. The plurality of batteries **280** are charged by the electric power generated by the photovoltaic unit **230**.

[0022] A flight control apparatus **262** and a communication control apparatus **264** are arranged in the central portion **260**. The flight control apparatus **262** controls flight of the HAPS **200**. The communication control apparatus **264** controls communication of the HAPS **200**.

[0023] The flight control apparatus **262** controls the flight of the HAPS **200** by controlling rotation of the propeller **240**, for example. In addition, the flight control apparatus **262** controls the flight of the HAPS **200** by changing an angle of the elevator **250**, for example. The flight control apparatus **262** may include various sensors such as a positioning sensor like a GPS sensor, a gyro sensor, an acceleration sensor, and a wind speed sensor and manage a position, an attitude, a movement direction, and a movement speed of the HAPS **200** and a wind speed around the HAPS **200**.

[0024] The communication control apparatus **264** forms the communication area **204** over the ground by using an SL (Service Link) antenna. The communication control apparatus **264** may form a service link with the user terminal **70** on the ground by using the SL antenna. The SL antenna may be a multi-beam antenna. The communication area **204** may be a multi-cell.

[0025] The communication control apparatus **264** may form a feeder link with a gateway **40** on the ground by using an FL (Feeder Link) antenna. The communication control apparatus **264** may access a network **30** via the gateway **40**.

[0026] The communication control apparatus **264** may communicate with a communication satellite **50** by using a satellite communication antenna. The communication control apparatus **264** may access the network **30** via the communication satellite **50** and a satellite communication station **60**.

[0027] The communication control apparatus **264** may communicate with the communication control apparatus **264** mounted to another HAPS **200** via the gateway **40** or the communication satellite **50** and the network **30**.

[0028] The user terminal **70** may be any communication terminal as long as it can communicate with the HAPS **200**. For example, the user terminal **70** is a mobile phone such as a smartphone. The user terminal **70** may also be a tablet terminal, a PC (Personal Computer), and the like. The

user terminal **70** may also be a so-called IoT (Internet of Things) device. The user terminal **70** can include anything that corresponds to so-called IoE (Internet of Everything).

[0029] The HAPS **200** relays a communication between the network **30** and the user terminal **70**, for example, via the feeder link or the communication satellite **50** and the service link. The HAPS **200** may provide a wireless communication service to the user terminal **70** by relaying the communication between the user terminal **70** and the network **30**.

[0030] The network **30** includes a mobile communication network. The mobile communication network may conform to any of an LTE (Long Term Evolution) communication system, a 5G (5th Generation) communication system, a 3G (3rd Generation) communication system, and a 6G (6th Generation) communication system and a communication system of the subsequent generation. The network **30** may include the Internet.

[0031] For example, the HAPS **200** transmits data received from the user terminal **70** in the communication area **204** to the network **30**. In addition, for example, when the HAPS **200** receives data addressed to the user terminal **70** in the communication area **204** via the network **30**, the HAPS **200** transmits the data to the user terminal **70**. The HAPS **200** may relay communication of another HAPS **200**. For example, the HAPS **200** directly wirelessly communicates with another HAPS **200** to transmit, via the network **30**, data received from the other HAPS **200** or transmit data received from the network **30** to the other HAPS **200**.

[0032] A management apparatus **400** manages a plurality of the HAPS **200**. The management apparatus **400** may communicate with the HAPS **200** via the network **30** and the gateway **40**. The management apparatus **400** may communicate with another HAPS **200** via the network **30**, the gateway **40**, and one of the HAPS **200**. The management apparatus **400** may communicate with the HAPS **200** via the network **30**, the satellite communication station **60**, and the communication satellite **50**.

[0033] The management apparatus **400** controls the HAPS **200** by transmitting an instruction. The management apparatus **400** may cause the HAPS **200** to circle over a target area such that the target area on the ground is covered by the communication area **204**. For example, the HAPS **200** maintains the feeder link with the gateway **40** by adjusting a pointing direction of the FL antenna while flying in a circular orbit over the target area and maintains a coverage of the target area by the communication area **204** by adjusting a pointing direction of the SL antenna.

[0034] FIG. **2** is an explanatory diagram for describing a power generation system **20** included in the HAPS **200**. The power generation system **20** exemplified in FIG. **2** includes a string **231** in which a plurality of power generation panels are connected and a power conditioner **232** which converts electric power generated by the string **231**, and further includes an electric storage unit **234** which is connected to the string **231** and stores the electric power generated by the string **231**. As exemplified in FIG. **2**, the power generation system **20** may include a plurality of combinations of the strings **231**, the power conditioners **232**, and the electric storage units **234**. Examples of the power conditioner **232** include an MPPT (Maximum Power Point Tracking).

[0035] The power generation system **20** may include a photovoltaic panel instead of the string **231**. Examples of the electric storage unit **234** include a capacitor, a super capacitor, a condenser, a battery, and the like.

[0036] The plurality of power conditioners **232** may be connected to a bus **290**. As exemplified in FIG. **2**, a PDU (Power Distribution Unit) **244**, a motor **242**, and the propeller **240**, an SDU (State Distribution Unit) **254**, a motor **252**, and the elevator **250**, and a battery **280** may be connected to the bus **290**. In addition, optional apparatuses such as a flight control apparatus, an antenna, and a light (none of which are illustrated in the drawing) may be connected to the bus **290**.

[0037] The HAPS **200** flies in a predetermined range of the stratosphere, for example, and the string **231** generates electric power by sunlight, but due to various factors, a power generation amount may be reduced. For example, at the time of sunrise or sunset, since solar radiation reaching the string **231** is weak, the power generation amount by the string **231** is reduced. In

addition, since the wing portion **220** of the HAPS **200** has a curved shape, even in a daytime, depending on the attitude of the HAPS **200**, the power generation amount by the string **231** may be reduced. In addition, the solar radiation reaching the string **231** is reduced as the distance of the flight area of the HAPS **200** from the vicinity of the equator increases. In addition, depending on a season, the solar radiation reaching the string **231** may be reduced.

[0038] Since the power conditioner **232** connected to the string **231** in this example supplies electric power to the motor **242**, the motor **252**, and the like, a relatively high output is required. The power conditioner **232** requires an input electric power at a certain level or above and cannot convert the input electric power into output electric power when the electric power amount is lower than a certain amount. In this manner, electric power less than minimum electric power that can be converted to output electric power by the power conditioner **232** may be described as weak electric power. When the electric power generated by the string **231** is the weak electric power, even when the string **231** generates electric power, the electric power ends up being wasted.

[0039] In contrast to the above, the power generation system **20** according to the present embodiment recovers the weak electric power by the electric storage unit **234**.

[0040] FIG. **3** schematically illustrates an example of an arrangement of the electric storage unit **234**. In the example illustrated in FIG. **3**, the electric storage unit **234** is connected in series to the string **231** and the power conditioner **232**. The electric power generated by the string **231** is stored in the electric storage unit **234**, and the electric power stored in the electric storage unit **234** is supplied to the power conditioner **232**.

[0041] In a case where the string **231** outputs weak electric power, when it is supplied to the power conditioner **232** as in related art, the weak electric power cannot be converted into the output electric power in the power conditioner **232**, and the weak electric power is wasted. In contrast to the above, according to the power generation system **20** illustrated in FIG. **3**, the electric power generated by the string **231** is stored in the electric storage unit **234** to become consolidated electric power which then flows to be supplied to the power conditioner **232**, so that waste of the weak electric power can be avoided.

[0042] In the example illustrated in FIG. **3**, the power conditioner **232** may control whether or not the electric power from the electric storage unit **234** is to be received. The power conditioner **232** may adjust, by a switch or the like, whether or not the electric power from the electric storage unit **234** is to be received. For example, the power conditioner **232** may acquire a measurement result of the power generation amount by the string **231** and perform control in a manner that the electric power is to be received from the electric storage unit **234** when the measurement result is higher than a predetermined threshold, and the electric power from the electric storage unit **234** is not to be received when the measurement result is lower than the threshold. The power conditioner **232** may acquire a measurement result by an ammeter arranged between the string **231** and the electric storage unit **234**, for example. By performing such control, for example, while the string **231** outputs the weak electric power, the weak electric power can be stored in the electric storage unit **234**, and when the string **231** outputs the electric power higher than the weak electric power, the electric power can be supplied to the power conditioner **232**.

[0043] FIG. **4** schematically illustrates an example of the arrangement of the electric storage unit **234**. In the example illustrated in FIG. **4**, the electric storage unit **234** is connected in parallel to connection between the string **231** and the power conditioner **232**. The electric power stored in the electric storage unit **234** is supplied to the power conditioner **232**.

[0044] As illustrated in FIG. **4**, by arranging the electric storage unit **234** to the connection between the string **231** and the power conditioner **232**, the weak electric power generated by the string **231** can be stored in the electric storage unit **234**.

[0045] FIG. **5** schematically illustrates an example of the arrangement of the electric storage unit **234**. In the example illustrated in FIG. **5**, the power generation system **20** includes a switch unit **236** and a switch control unit **237**.

[0046] The switch unit **236** may be a switch which switches the current from the string **231** to be supplied to the power conditioner **232** or to be supplied to the electric storage unit **234**. As exemplified in FIG. **4**, the switch unit **236** may be arranged at a position branching from the connection between the string **231** and the power conditioner **232** to the electric storage unit **234**. The switch unit **236** may be arranged at another position along a way to the connection between the string **231** and the electric storage unit **234**.

[0047] By controlling the switch unit **236**, the switch control unit **237** switches the current of the string **231** to be supplied to the power conditioner **232** or to be supplied to the electric storage unit **234**. In the example illustrated in FIG. **5**, the switch control unit **237** acquires a measurement result of the power generation amount by the string **231** and switches the switch unit **236** in a manner that the current from the string **231** is to be supplied to the power conditioner **232** when the measurement result is higher than a predetermined threshold or the current from the string **231** is to be supplied to the electric storage unit **234** when the measurement result is lower than the threshold. The threshold may be a value with which it can be determined on whether or not the measurement result corresponds to a weak current. The switch control unit **237** may acquire a measurement result by an ammeter arranged between the string **231** and the switch unit **236**, for example.

[0048] When the switch control unit **237** performs such switching, in a case where the electric power generated by the string **231** is the weak electric power, all the weak electric power can be stored in the electric storage unit **234**, and the weak electric power can be effectively used. In addition, when the sufficient electric power is generated by the string **231**, it is possible to cause the current to flow through the power conditioner **232** without causing the current to flow through the electric storage unit **234**, and life of the electric storage unit **234** can be extended as compared to a case where the switching is not performed.

[0049] FIG. **6** schematically illustrates an example of the arrangement of the electric storage unit **234**. Here, differences from FIG. **5** will be mainly described. In the example illustrated in FIG. **6**, the power generation system **20** further includes an information acquisition unit **238**.

[0050] The information acquisition unit **238** acquires flight related information in connection to the flight of the HAPS **200**. The flight related information may include a time period in which the HAPS **200** flies.

[0051] The information acquisition unit **238** may acquire the flight related information from the flight control apparatus **262**. The flight related information includes, for example, a flight position of the HAPS **200**. When the HAPS **200** flies on a particular path, the flight related information includes, for example, a flight path of the HAPS **200**. The flight related information includes, for example, an attitude of the HAPS **200**. The flight related information includes, for example, a movement direction of the HAPS **200**. The flight related information includes, for example, a movement speed of the HAPS **200**. The flight related information includes, for example, a wind speed around the HAPS **200**. The flight related information includes, for example, a solar altitude. The flight related information includes, for example, a solar mid-sun altitude. The flight related information includes, for example, an intensity distribution of light at a flight altitude or a flight point. The flight related information includes, for example, a reflective power at the flight altitude or the flight point. The flight related information includes, for example, a degree of scattering of light at the flight altitude or the flight point. The flight related information includes, for example, an output current amount from the string **231**. The flight related information includes, for example, an output voltage from the string **231**. In this manner, the flight related information may include a state of energy.

[0052] The information acquisition unit **238** may acquire the flight related information from the communication control apparatus **264**. The flight related information may include information related to the flight area which is received by the flight control apparatus **262** from another HAPS **200**. The flight control apparatus **262** receives wind information including a wind direction and a

wind speed of the flight area, for example, from another HAPS **200** flying ahead of its own flying vehicle on a same flight path as the HAPS **200** to which the flight control apparatus **262** is mounted (which may be described as its own flying vehicle) and provides the wind information to the information acquisition unit **238**.

[0053] The switch control unit **237** switches the switch unit **236** based on the flight related information acquired by the information acquisition unit **238** in a manner that the current from the string **231** is to be supplied to the power conditioner **232** when a state in which the power generation amount by the string **231** is high is estimated or the current from the string **231** is to be supplied to the electric storage unit **234** when a state in which the power generation amount by the string **231** is low is estimated.

[0054] For example, when a time period in which the HAPS **200** flies is a predetermined time period set as a time period of sunrise, the switch control unit **237** switches the current from the string **231** to be supplied to the electric storage unit **234**. For example, when the time period in which the HAPS **200** flies is a predetermined time period set as a time period of sunset, the switch control unit **237** switches the current from the string **231** to be supplied to the electric storage unit **234**. With these configurations, when an amount of solar radiation reaching the string **231** is low and a power generation amount by the string **231** becomes the weak electric power during the time period of sunrise or the time period of sunset, by supplying the weak electric power to the power conditioner **232**, waste of the weak electric power can be avoided, and by supplying the weak electric power to the electric storage unit **234**, the weak electric power can be stored.

[0055] For example, the switch control unit **237** estimates a power generation amount by the string **231** based on a flight path of the HAPS **200** and a power generation amount of the string **231** when the HAPS **200** flew on the flight path in a past, and the current from the string **231** is to be supplied to the electric storage unit **234** when the estimated power generation amount is lower than a predetermined threshold, or the current from the string **231** is to be supplied to the power conditioner **232** when the estimated power generation amount is higher than the threshold.

[0056] For example, the switch control unit **237** estimates a gradient of its own flying vehicle to which the switch control unit **237** is mounted based on wind information acquired from another HAPS **200** which flies ahead of its own flying vehicle on a same flight path as its own flying vehicle, estimates a power generation amount by the string **231** based on the estimated gradient of its own flying vehicle, and the current from the string **231** is to be supplied to the electric storage unit **234** when the estimated power generation amount is lower than a predetermined threshold, or the current from the string **231** is to be supplied to the power conditioner **232** when the estimated power generation amount is higher than the threshold.

[0057] In the HAPS **200**, the wing portion **220** may be transparent, and the photovoltaic unit **230** may have a bifacial photovoltaic panel which can generate electric power on both a front surface and a back surface. In addition, the HAPS **200** may include a front side photovoltaic unit arranged on a front side of the wing portion **220** and a back side photovoltaic unit arranged on a back side of the wing portion **220**. In this case, the switch control unit **237** may switch, for each of the front side photovoltaic unit and the back side photovoltaic unit, the generated current to be supplied to the power conditioner **232** or to be supplied to the electric storage unit **234** based on the flight related information. For example, when the sun is positioned on an upper surface side of the HAPS **200**, a power generation amount of the front side photovoltaic unit becomes high, and a power generation amount of the back side photovoltaic unit becomes low. In addition, for example, when the sun is positioned on an lower surface side of the HAPS **200**, the power generation amount of the front side photovoltaic unit becomes low, and the power generation amount of the back side photovoltaic unit becomes high. In this manner, in the front side photovoltaic unit and the back side photovoltaic unit, tendencies of the power generation amounts are different depending on a situation of the HAPS **200**. Therefore, for each of the front side photovoltaic unit and the back side photovoltaic unit, by performing the switching based on the flight related information in a way that suits each of

those, a recovery rate of the weak electric power can be increased as compared to a case where no distinction is made between the front side photovoltaic unit and the back side photovoltaic unit. [0058] FIG. 7 schematically illustrates an example of the arrangement of the electric storage unit 234. In the example illustrated in FIG. 7, the electric storage unit 234 is connected to the connection between the string 231 and the power conditioner 232, and the electric power stored in the electric storage unit 234 is supplied to a system other than the power conditioner 232. In the example illustrated in FIG. 7, the electric power stored in the electric storage unit 234 is supplied to a power conditioner for weak electric power 235. For example, the electric power converted by the power conditioner 232 from the electric power generated by the string 231 is supplied to the motor 242 and the motor 252 included in the HAPS 200, and the electric power supplied to the power conditioner for weak electric power 235 is supplied to a communication related apparatus which can operate even with lower electric power compared to the motor 242 and the motor 252. Examples of the communication related apparatus include the communication control apparatus 264, a positioning sensor, and the like.

[0059] FIG. 8 schematically illustrates an example of the arrangement of the electric storage unit 234. Here, differences from FIG. 7 will be mainly described. In the example illustrated in FIG. 8, the power generation system 20 includes the switch unit 236 and the switch control unit 237. The switch unit 236 and the switch control unit 237 may be similar to the switch unit 236 and the switch control unit 237 of FIG. 5.

[0060] FIG. 9 schematically illustrates an example of the arrangement of the electric storage unit 234. Here, differences from FIG. 8 will be mainly described. In the example illustrated in FIG. 9, the power generation system 20 further includes the information acquisition unit 238. The information acquisition unit 238 may be similar to the information acquisition unit 238 of FIG. 6.

[0061] FIG. 10 schematically illustrates an example of the arrangement of the electric storage unit 234. Here, differences from FIG. 7 will be mainly described. The electric storage unit 234 may be connected to each of connections between a plurality of strings 231 and a plurality of power conditioners 232, and the electric power stored in the electric storage unit 234 may be supplied to a system other than the power conditioner 232. FIG. 10 illustrates an example in which the single electric storage unit 234 is arranged for two combinations of the strings 231 and the power conditioners 232, but the configuration is not limited to this, and the single electric storage unit 234 may be arranged for three or more combinations of the strings 231 and the power conditioners 232. In this manner, by arranging the single electric storage unit 234 for a plurality of combinations of the strings 231 and the power conditioners 232, it is possible to reduce a total weight and costs as compared to a case where the electric storage unit 234 is arranged for each of the plurality of combinations of the strings 231 and the power conditioners 232.

[0062] FIG. 11 schematically illustrates an example of a hardware configuration of a computer 1200 which functions as a control apparatus including the switch control unit 237 and the information acquisition unit 238. Programs installed in the computer 1200 can cause the computer 1200 to function as one or more “units” of the apparatus according to the embodiment described above or can cause the computer 1200 to execute operations associated with the apparatuses according to the embodiment described above or the one or more “units”, and/or can cause the computer 1200 to execute a process according to the embodiment described above or stages of the process. Such a program may be executed by a CPU 1212 to cause the computer 1200 to perform particular operations associated with some or all of the blocks in the flowcharts and block diagrams described in the present specification.

[0063] The computer 1200 according to the present embodiment includes the CPU 1212, a RAM 1214, and a graphics controller 1216, which are connected to each other via a host controller 1210. Also, the computer 1200 includes input/output units such as a communication interface 1222, a storage device 1224, a DVD drive and an IC card drive, which are connected to the host controller 1210 via an input/output controller 1220. The storage device 1224 may be a hard disk drive, a

solid-state drive, and the like. The computer **1200** also includes a ROM **1230** and a legacy input/output unit such as a keyboard, which are connected to the input/output controller **1220** via an input/output chip **1240**.

[0064] The CPU **1212** operates in accordance with the programs stored in the ROM **1230** and the RAM **1214**, thereby controlling each unit. The graphics controller **1216** acquires image data which is generated by the CPU **1212** in a frame buffer or the like provided in the RAM **1214** or in itself so as to cause the image data to be displayed on a display device **1218**.

[0065] The communication interface **1222** communicates with other electronic devices via a network. The storage device **1224** stores a program and data used by the CPU **1212** in the computer **1200**. The IC card drive reads the program and data from an IC card, and/or writes the program and data to the IC card.

[0066] The ROM **1230** stores therein a boot program or the like executed by the computer **1200** at the time of activation, and/or a program depending on the hardware of the computer **1200**. The input/output chip **1240** may also connect various input/output units via a USB port, a parallel port, a serial port, a keyboard port, a mouse port, or the like to the input/output controller **1220**.

[0067] A program is provided by a computer readable storage medium such as the DVD-ROM or the IC card. The program is read from the computer readable storage medium, installed into the storage device **1224**, RAM **1214**, or ROM **1230**, which are also examples of a computer readable storage medium, and executed by the CPU **1212**. Information processing written in these programs is read by the computer **1200**, and provides cooperation between the programs and the various types of hardware resources described above. An apparatus or method may be configured by implementing the operation or processing of information in accordance with the usage of the computer **1200**.

[0068] For example, when a communication is performed between the computer **1200** and an external device, the CPU **1212** may execute a communication program loaded in the RAM **1214** and instruct the communication interface **1222** to perform communication processing based on a process written in the communication program. The communication interface **1222**, under control of the CPU **1212**, reads transmission data stored on a transmission buffer region provided in a recording medium such as the RAM **1214**, the storage device **1224**, the DVD-ROM, or the IC card, and transmits the read transmission data to a network or writes reception data received from a network to a reception buffer region or the like provided on the recording medium.

[0069] In addition, the CPU **1212** may cause all or a necessary portion of a file or a database to be read into the RAM **1214**, the file or the database having been stored in an external recording medium such as the storage device **1224**, the DVD drive (DVD-ROM), the IC card, etc., and perform various types of processing on the data on the RAM **1214**. Next, the CPU **1212** may write the processed data back into the external recording medium.

[0070] Various types of information, such as various types of programs, data, tables, and databases, may be stored in the recording medium to undergo information processing. The CPU **1212** may execute, on the data read from the RAM **1214**, various types of processing including various types of operations, information processing, conditional judgement, conditional branching, unconditional branching, information search/replacement, or the like described throughout the present disclosure and designated by instruction sequences of the programs, to write the results back to the RAM **1214**. In addition, the CPU **1212** may search for information in a file, a database, or the like in the recording medium. For example, when a plurality of entries, each having an attribute value of a first attribute associated with an attribute value of a second attribute, are stored in the recording medium, the CPU **1212** may search for an entry whose attribute value of the first attribute matches a designated condition, from among the said plurality of entries, and read the attribute value of the second attribute stored in the said entry, thereby acquiring the attribute value of the second attribute associated with the first attribute that satisfies a predetermined condition.

[0071] The programs or software module described above may be stored on the computer **1200** or

in a computer readable storage medium near the computer **1200**. In addition, a recording medium such as a hard disk or a RAM provided in a server system connected to a dedicated communication network or the Internet can be used as the computer readable storage medium, thereby providing the program to the computer **1200** via the network.

[0072] Blocks in flowcharts and block diagrams in the present embodiments may represent stages of processes in which operations are executed or “units” of apparatuses responsible for executing operations. A specific step and “unit” may be implemented by a dedicated circuit, a programmable circuit supplied along with a computer readable instruction stored on a computer readable storage medium, and/or a processor supplied along with the computer readable instruction stored on the computer readable storage medium. The dedicated circuit may include a digital and/or analog hardware circuit, or may include an integrated circuit (IC) and/or a discrete circuit. The programmable circuit may include, for example, a reconfigurable hardware circuit including logical AND, logical OR, logical XOR, logical NAND, logical NOR, and another logical operation, and a flip-flop, a register, and a memory element, such as a field programmable gate array (FPGA), a programmable logic array (PLA), or the like.

[0073] The computer readable storage medium may include any tangible device capable of storing an instruction executed by an appropriate device, so that the computer readable storage medium having the instruction stored thereon constitutes a product including an instruction that may be executed in order to provide means for executing an operation designated by a flowchart or a block diagram. An example of the computer readable storage medium may include an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, etc. More specific examples of computer readable storage medium may include a floppy disk, a diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an electrically erasable programmable read-only memory (EEPROM), a static random access memory (SRAM), a compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a BLU-RAY (registered trademark) disc, a memory stick, an integrated circuit card, etc.

[0074] The computer readable instructions may include an assembler instruction, an instruction-set-architecture (ISA) instruction, a machine instruction, a machine-dependent instruction, a microcode, a firmware instruction, state-setting data, or either of source code or object code written in any combination of one or more programming languages including an object-oriented programming language such as SMALLTALK (registered trademark), JAVA (registered trademark), and C++, or the like, and a conventional procedural programming language such as a “C” programming language or a similar programming language.

[0075] The computer readable instruction may be provided to a general purpose computer, a special purpose computer, or a processor or programmable circuit of another programmable data processing apparatus locally or via a local area network (LAN), a wide area network (WAN) such as the Internet or the like in order that the general purpose computer, the special purpose computer, or the processor or the programmable circuit of another programmable data processing apparatus executes the computer readable instruction to generate means for executing operations designated by the flowchart or the block diagram. Examples of the processor include a computer processor, a processing unit, a microprocessor, a digital signal processor, a controller, a microcontroller, and the like.

[0076] While the present invention has been described by way of the embodiments, the technical scope of the present invention is not limited to the scope described in the above-described embodiments. It is apparent to persons skilled in the art that various alterations or improvements can be made to the above-described embodiments. It is also apparent from the described scope of the claims that the embodiments added with such alterations or improvements can be included the technical scope of the present invention.

[0077] It should be noted that the operations, procedures, steps, stages or the like of each process performed by an apparatus, system, program, and method shown in the scope of the claims, embodiments, or diagrams can be performed in any order as long as the order is not indicated by “prior to,” “before,” or the like and as long as the output from a previous process is not used in a later process. Even if the process flow is described by using phrases such as “first”, “then” or the like in the scope of the claims, embodiments, or diagrams, it does not necessarily mean that the process must be performed in this order.

[0078] With use of the present invention, a power generation efficiency per weight of the solar cell can be improved, and the solar cell that is lighter and can generate more electric power can be achieved. Since it is conceivable to use the solar cell to be brought to a disaster site, for example, to supply energy to victims or the like, it is possible to achieve Sustainable Development Goals (SDGs) such as Goal 7 “Affordable and clean energy”, Goal 13 “Take specific action to combat climate change”, or the like.

EXPLANATION OF REFERENCES

[0079] **20**: power generation system; **30**: network; **40**: gateway; **50**: communication satellite; **60**: satellite communication station; **70**: user terminal; **200**: HAPS; **202**: beam; **204**: communication area; **220**: wing portion; **230**: photovoltaic unit; **231**: string; **232**: power conditioner; **234**: electric storage unit; **235**: power conditioner for weak electric power; **236**: switch unit; **237**: switch control unit; **238**: information acquisition unit; **240**: propeller; **242**: motor; **244**: PDU; **250**: elevator; **252**: motor; **254**: SDU; **260**: central portion; **262**: flight control apparatus; **264**: communication control apparatus; **270**: pod; **280**: battery; **290**: bus; **400**: management apparatus; **1200**: computer; **1210**: host controller; **1212**: CPU; **1214**: RAM; **1216**: graphics controller; **1218**: display device; **1220**: input/output controller; **1222**: communication interface; **1224**: storage device; **1230**: ROM; and **1240**: input/output chip.

Claims

1. A power generation system comprising: a photovoltaic unit arranged in a flight vehicle which provides a wireless communication service to a user terminal in a communication area formed over a ground; an electric storage unit which is connected to the photovoltaic unit and stores electric power generated by the photovoltaic unit; and a power conditioner which converts the electric power generated by the photovoltaic unit, wherein the electric storage unit is connected in parallel to connection between the photovoltaic unit and the power conditioner, and the electric power stored in the electric storage unit is supplied to the power conditioner.
2. The power generation system according to claim 1, further comprising: a switch control unit which switches a current from the photovoltaic unit to be supplied to the power conditioner or to be supplied to the electric storage unit.
3. The power generation system according to claim 2, wherein the switch control unit switches the current from the photovoltaic unit to be supplied to the power conditioner when a measurement result of a power generation amount by the photovoltaic unit is higher than a predetermined threshold or the current from the photovoltaic unit to be supplied to the electric storage unit when the measurement result is lower than the threshold.
4. The power generation system according to claim 2, further comprising: an information acquisition unit which acquires flight related information in connection to flight of the flight vehicle, wherein the switch control unit switches the current from the photovoltaic unit to be supplied to the power conditioner or to be supplied to the electric storage unit based on the flight related information.
5. The power generation system according to claim 4, wherein the switch control unit switches, based on the flight related information, the current from the photovoltaic unit to be supplied to the power conditioner when a state in which a power generation amount by the photovoltaic unit is

high is estimated or the current from the photovoltaic unit to be supplied to the electric storage unit when a state in which a power generation amount by the photovoltaic unit is low is estimated.

6. The power generation system according to claim 4, wherein the switch control unit switches the current from the photovoltaic unit to be supplied to the electric storage unit when a time period in which the flight vehicle flies is a predetermined time period set as a time period of sunset.

7. The power generation system according to claim 4, wherein the flight vehicle has a front side photovoltaic unit arranged on a front side of a wing portion of the flight vehicle and a back side photovoltaic unit arranged on a back side of the wing portion, and the switch control unit switches, for each of the front side photovoltaic unit and the back side photovoltaic unit, a generated current to be supplied to the power conditioner or to be supplied to the electric storage unit based on the flight related information.

8. A power generation system comprising: a photovoltaic unit arranged in a flight vehicle which provides a wireless communication service to a user terminal in a communication area formed over a ground; an electric storage unit which is connected to the photovoltaic unit and stores electric power generated by the photovoltaic unit; and a power conditioner which converts the electric power generated by the photovoltaic unit, wherein the electric storage unit is connected in series to the photovoltaic unit and the power conditioner, and the electric power stored in the electric storage unit is supplied to the power conditioner.

9. A power generation system comprising: a photovoltaic unit arranged in a flight vehicle which provides a wireless communication service to a user terminal in a communication area formed over a ground; an electric storage unit which is connected to the photovoltaic unit and stores electric power generated by the photovoltaic unit; and a power conditioner which converts the electric power generated by the photovoltaic unit, wherein the electric storage unit is connected to connection between the photovoltaic unit and the power conditioner, and the electric power stored in the electric storage unit is supplied to a system other than the power conditioner.

10. The power generation system according to claim 9, wherein the photovoltaic unit includes a plurality of photovoltaic units, the power conditioner includes a plurality of power conditioners, the electric storage unit is connected to each of connections between the plurality of photovoltaic units and the plurality of power conditioners, and the electric power stored in the electric storage unit is supplied to the system other than the power conditioner.

11. The power generation system according to claim 9, wherein the electric power converted by the power conditioner from the electric power generated by the photovoltaic unit is supplied to a motor included in the flight vehicle, and the electric power stored in the electric storage unit is supplied to a communication related apparatus in connection to communication by the flight vehicle.

12. A flight vehicle that is the flight vehicle comprising the power generation system according to any one of claims 1 to 11, the flight vehicle further comprising: a flight control apparatus which controls the flight of the flight vehicle by using the electric power converted by the power conditioner.

13. A flight vehicle which provides a wireless communication service to a user terminal in a communication area formed over a ground, the flight vehicle comprising: a power generation system including a photovoltaic unit arranged in the flight vehicle, an electric storage unit which is connected to the photovoltaic unit and stores electric power generated by the photovoltaic unit, and a power conditioner which converts the electric power generated by the photovoltaic unit; and a flight control apparatus which controls the flight of the flight vehicle by using the electric power converted by the power conditioner.
